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None.

Sex hormones and acne: State of the art

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Summary

Acne is an androgen-dependent inflammatory disease of sebaceous follicles. Herein, we reviewed and discussed the underlying pathways of androgen biosynthesis and metabolism, non-genomic regulation of androgen receptor expression and function, posttranslational regulation of androgen excess in acne and acne-associated syndromes, such as polycystic ovary syndrome, and congenital adrenal hyperplasia. We provide insights into the involvement of sex hormones, particularly androgens, in skin homeostasis and acne pathogenesis, including comedogenesis, lipogenesis, microbiota, and inflammation. Advanced understanding of the action mechanisms of classical acne treatment and new development of antiandrogens, both topical and systemic, are also highlighted.

Introduction

Strong research evidence shows that acne is an androgen-dependent, insulin-like growth factor (IGF)-1-driven disease of sebaceous follicles [1, 2]. In this regard, the interactions between insulin/IGF-1, androgen synthesis, and p53-mediated signaling pathways are crucial in the pathogenesis of acne [2]. Both testosterone and dihydrotestosterone (DHT) drive the androgen receptor (AR)-dependent genomic effect in sebocytes, where the non-genomic pathway is little studied [3]. The clinical significance of the backdoor biosynthesis of DHT bypassing testosterone remains elusive. Androgen-mediated lipid synthesis and inflammation have been reported in sebocytes [3], while its role in comedogenesis remains unclear. Studies have reported that genetic susceptibility and polymorphism associated with AR is involved in severe acne [3]. In the treatment of acne, anti-androgens, including combined oral contraceptives (COCs), are widely used, however, without strong evidence on their efficacy [4]. In this review, we evaluate and discuss important findings reported in the past three years, focused on androgen physiology, the effects on sebocytes, and acne pathogenesis, as well as novel therapeutics in development.

Androgen biosynthesis

The alternative (backdoor) androgen synthesis pathway has been reported in human hyperandrogenic disorders, such as prenatal virilization/masculinization in girls with congenital adrenal hyperplasia (CAH), or with prepubertal acne [5]. Studies have demonstrated that androsterone is the principal backdoor androgen in the human fetus, and its circulating level is sex-dependent. The backdoor androgens are primarily synthesized from the placental progesterone in several nontesticular tissues, with little *de novo* biosynthesis in the testis. This explains the relationship between placental insufficiency and sex development disorders [5]. The expression of aldo-keto reductases (AKR1C1-1C4), 5α-reductases (SRD5A1/2), and retinol dehydrogenase (RoDH) in the ovarian theca cells is enhanced in the polycystic ovary syndrome (PCOS), revealing that backdoor DHT biosynthesis occurs in the ovary and plays significant roles in androgen overproduction and clinical hyperandrogenism (HA) [6].

Intracrine androgen synthesis is often not reflected by circulating androgens but rather by androgen metabolites and conjugates [7]. The clinical mass spectrometry applied in steroid profiling highlights the significance of several previously neglected metabolites, particularly the adrenal-derived 11-oxygenated 19-carbon (11oxC19) steroids originating from the adrenal cortex, which are elevated in circulation in patients with 21-hydroxylase deficiency (21-OHD) and PCOS [8]. The 11-ketotestosterone and its 5α-reduced metabolite 11-ketodihydrotestosterone, similar to testosterone and DHT, respectively, are potent AR agonists [8].

A recent *in vitro* study showed that nicotinamide potentially inhibited testosterone biosynthesis in the human adrenal NCI-H295R cells, and regulated the expressions of critical enzymes in the androgen biosynthesis pathway, as well as the genes related to steroid biosynthesis, TGFβ signaling, and the targets of IGF-1/IGF-2 [9]. In human sebocytes, nicotinic acid has been found to suppress lipogenesis via activation of the hydroxycarboxylic acid receptor 2 *in vitro* [10]. Well controlled clinical trials, especially regarding the dose-response effect, are required to prove its potential in acne treatment.

Androgen metabolism in PCOS and CAH

In a retrospective study, excess androgen levels were reported in 199 out of 487 children, with 140 at pre-pubertal and 59 at post-pubertal age [11]. In the pre-pubertal age, premature adrenarche was the most common diagnosis (61 %), characterized by dehydroepiandrosterone sulfate (DHEAS) excess in 85 %. In the post-pubertal age, PCOS was diagnosed in 40 % of the children, with equally frequent isolated excess of DHEAS (29 %) or testosterone (25 %), or increases in both androstenedione and testosterone (25 %).

In addition to genetic factors, studies reported that post-transcriptional regulation via small noncoding microRNAs (miRNAs) is associated with the androgen excess in PCOS [12], in which hyperandrogenemia is linked to serum-derived miRNAs, including miR-1290, miR-139-3p, and miR361-5p, whereas miR-361-5p, miR1225-3p, and miR-34-3p are correlated with the metabolic syndrome [12]. MiR-135a, an androgen-regulated miRNA, is upregulated in PCOS, where it represses the proliferation and induces apoptosis of the granulosa cells (GCs) by downregulating vascular endothelial growth factor C *in vitro* [13]. The upregulation of miR-204 can reduce testosterone levels, enhance cell proliferation, and suppress cell apoptosis in human GCs, whereas the overexpression of miR-3940-5p promotes cell proliferation *in vitro* [14]. In PCOS ovaries, the miRNAs modulating insulin sensitivity consisting of miR-146a, miR-155, miR-204, miR-320, miR-370, miR-486, and miR-612 have been found to be dysregulated. This supports the pathogenic role of insulin resistance in PCOS [15].

Long non-coding RNAs (lncRNAs), including the highly up-regulated in PCOS (HUPCOS), HLA complex P5 (HCP5), and the metastasis-associated lung adenocarcinoma transcript 1 (MALAT1) are critically involved in the regulation of the proliferation, apoptosis, and androgen biosynthesis in human ovarian GCs [16–18]. An *in vitro* study showed that a subset of miRNA (overexpressed miR-203 and miR-574-3p, and downregulated miR-7) modulates sebaceous lipogenesis [19]. The role of miRNAs or lncRNAs in androgen biosynthesis in human sebocytes has not been elucidated.

Preliminary findings showed that the level of DNA methylation of the fermitin family homolog 2 (FERMT2), a gene that codes for the scaffold protein related to AR, was higher in women with PCOS compared with the healthy population [20]. There was no significant difference in FERMT2 levels in PCOS patients with oligomenorrhea or regular cycles, indicating that the two sub-phenotypes are epigenetically similar.

AR and the skin

The AR functions through genomic and non-genomic signaling pathways [3]. A recent study showed that the tyrosine kinase-dependent pathway regulates AR [21]. In the LNCaP prostate cancer cells, phosphorylation of AR at Tyr-267 by the Ack1/TNK2 tyrosine kinase led to nuclear translocation, DNA binding, and androgen-dependent gene transcription in a low androgen environment. In the absence of androgen, a downstream gene of the regulatory pathway referred to as the *SRA stem-loop interacting RNA binding protein* (*SLIRP*) binds to the androgen response elements of the AR target genes. Whether this pathway exists in sebocytes and is related to acne is unclear.

Recently, it was suggested that mitochondria are novel players in the nongenomic activities of AR [22]. It was shown that in addition to nuclear localization, AR was imported into the mitochondria, where AR inversely regulated the expression, assembly, integrity, and functions of the mitochondrial complexes. Mitochondrial stress increased both AR expression and its translocation to the mitochondria. In human skin, an *ex vivo* culture model did not show any influence of testosterone or DHT on the major organelles including mitochondria, in the keratinocytes of different epidermal strata [23]. An *in vitro* study on human sebocytes demonstrated transcriptional changes in a mitochondria related pathway associated with age-specific decline in sex hormones [24]. Studies should be conducted to explore the presence and the effect of mitochondrial AR on overall mitochondrial function, as well as mitonuclear communication in the skin.

Androgen and comedogenesis

The action of androgens on comedogenesis was recently demonstrated *in vitro*, where the expression of growth factors in dermal fibroblasts was shown to regulate keratinocyte differentiation [25]. Androgens (testosterone and DHT), but not estrogens, promote the biosynthesis of growth factors, including amphiregulin, epiregulin, fibroblast growth factor 10 (FGF10), and IGF binding protein 5, in dermal fibroblasts, and not in keratinocytes. FGF10 suppresses the expression of CK1/CK10 and CK5 in keratinocytes. Similarly, DHT inhibits the expression of CK1 and CK10 in keratinocytes co-cultured with fibroblasts, but not in keratinocyte monocultures. This suggests that DHT modulates keratinocyte differentiation through the regulation of dermal fibroblasts. Increased staining signals of growth factors have been revealed in the dermis close to the hair infundibulum in acne lesions through laser scanning confocal microscopy, consistent with the altered CK10/CK14 ratio during keratinocyte differentiation in the hair infundibulum.

The aryl hydrocarbon receptor (AHR) is enriched in skin cells, especially in keratinocytes and sebocytes. Activation of these AHR by dioxins hyperactivates epidermal terminal differentiation (keratinization), converting sebocytes into keratinocytes, resulting in chloracne [26]. Further studies are required to investigate whether this signaling pathway is associated with the abnormal keratinization of the hair infundibulum and how it interacts with androgens in acne.

Androgen and skin microbiota

Little is known about the effect of androgens on skin microbiota. In patients with androgenetic alopecia, the causal association between the increased abundance of *Propionibacterium acnes* and miniaturized hair follicles is unclear [27]. An analysis of the fecal microbiota and serum levels of testosterone and E2 in males and females showed significant gender differences and a close correlation between the sex steroid hormone levels and the diversity and composition of gut microbiota [28]. It has also been shown that E2 is the primary sex hormone that drives the shifting of the vaginal microbiome towards *Lactobacillus* dominance in shaping the composition of the vaginal microbiome in pubertal girls compared with women at menopause [29]. In a rat model of PCOS, the exposure to androgens *in utero* causes dysbiosis of the intestinal microbiota and metabolic disorders of newborn female but not male rats [30]. On the other hand, in young, healthy mice and men, the gastrointestinal microbiota in the cecum and colon influences the metabolism and deglucuronidation of DHT and testosterone, resulting in extremely high free levels of unconjugated DHT in the colon [31]. The androgen effect on *P. acnes* and the potential androgen metabolism by *P. acnes* are largely unknown.

In a recent study, it was reported that stress meditators, including catecholamines (epinephrine and norepinephrine), increase biofilm formation of two subtypes of *P. acnes*, acneic RT4 and non-acneic RT6 strains without affecting proliferation [32]. *P. acnes* promotes the stimulating effect of catecholamines on lipid biosynthesis in sebocytes *in vitro*. This supports an interaction between stress and immune response in acne. Further studies should be conducted to probe the interaction of androgens with catecholamines in inflammation in acne.

Androgen and inflammation

Despite the widely accepted proinflammatory effects of androgens (DHT and testosterone) on sebocytes [3], sex hormones regulate proinflammatory cytokines in a complex manner: lower physiological levels of estrogens enhance the secretion of IL-12, IL-6, and IL-1β, which are inhibited at higher levels, along with promoted regulatory cytokines like IL-10 [33]. Testosterone reduces the expression of cytokines (IFNγ and TNFα), as well as diminishes the TLR-mediated responses of macrophages while increasing the expression of IL-10 in monocytes [34]. The ability of testosterone to suppress inflammation has been further shown in gonadectomy or hormone replacement in animal models [34]. A recent study documented the anti-inflammatory effect of androgens in male individuals, in which testosterone negatively regulates type 2 immune responses sustained by $CD4^+$ Th2 and type 2 innate lymphoid cells (ILC2), diminishes allergic airway inflammation and downregulates the expression of IL-17A protein secreted by $CD4^+$ Th17 cells in asthma [35]. In obese patients with acanthosis nigricans after laparoscopic sleeve gastrectomy, an improved inflammatory state associated with increased testosterone levels was reported [36]. ILC2 progenitors express specifically the AR but not the estrogen receptor, and AR activation directly inhibits the differentiation and proliferation of mature ILC2s [37]. In Behçet's disease, in contrast, testosterone hyperactivates neutrophils and alters the Th1 type immune response constituting an elevation of IL-12 and downregulation of IL-10, TLR4, ERAP1, and CCR1. This explains the more severe disease course and higher mortality among male patients [38]. Further studies should be conducted to explore whether and how androgens exert pro- or anti-inflammation effects in acne pathogenesis, especially the innate immune responses.

Estrogen, progesterone, prolactin and acne

Cytochrome (CYP) 19 is involved in estrogen synthesis. A recent study showed that the GG genotype of CYP19 rs700518 polymorphism increases the risk and severity of acne in females [39]. On the other hand, prolactin is considered as a sebotrophic hormone because of its direct and indirect effects on the sebaceous gland [40]. With the expression of the prolactin receptor in sebocytes, prolactin directly increases the human sebaceous gland area and sebocyte growth while indirectly regulating androgen metabolism via stimulation of lipid production and DHT biosynthesis in sebocytes by increasing levels of the 5α reductase 1 protein. In contrast, males with hyperprolactinemia have shown reduced serum testosterone levels, whereas prolactin represses 5α reductase activity in the genital skin derived from normoprolactinaemic non-hirsute women. Estrogen is supposed to act as an anti-androgenic and sebosuppressor, but the evidence is largely lacking [41]. The oxidative and reductive 17β-hydroxysteroid dehydrogenases (17βHSDs) interconvert 17-keto and 17β-hydroxysteroids; therefore, they determine the ultimate androgenic/estrogenic activity, which is crucial in intracrine metabolism of estrogens [42]. The androgenic 17βHSD3, which is generally considered testis-specific, is also expressed in both the subcutaneous and visceral fat tissue in women, indicating that the adipose tissue in women is substantially androgenic [42].

Androgens and acne

In a study involving 274 adolescents and young adults with PCOS aged between 13.0 and 24.9 years, the three PCOS phenotypes established in early adolescence remained constant into adulthood independent of BMI [43]. The risk of obesity was higher in young adults compared with adolescents. However, the risk of acne was higher in adolescents relative to young adults. In a study constituting 72 female patients with refractory acne, significantly higher levels of testosterone, DHEAS, prolactin, and LH were reported in patients aged 23–36 years than those aged 15–22 years. This suggested more evident hormone disturbance in adult acne. In another study involving 120 female acne patients aged ≥ 25 years, with 68 cases of late-onset and 52 cases of persistent acne, 71.67 % of cases exhibited clinical HA while 18.33 % had biochemical hyperandrogenemia [44]. A higher percentage of clinical HA, hormonal abnormalities, and PCOS was reported in the persistent acne patients compared with the late-onset population. Polycystic ovary syndrome was more frequent in persistent acne patients with clinical HA and increased 17-hydroxyprogesterone levels.

In a study of male hormonal contraceptives with high-dose testosterone or a combination of reduced testosterone with a progestin, acne was one of the most common side effects [45]. In a previous study with 62 adolescents of testosterone-treated gender dysphoria, acne was reported as a common side effect, at an incidence rate of 35–49 % and most prevalent at 6–12 months of testosterone therapy [46]. These results provide further evidence of the induction of acne by exogenous androgens.

The use of external androgens or anabolic steroids in the induction of acne has remained a hot topic in dermatology [47]. Recent studies provided further evidence of androgenic xenobiotics, especially in milk of non-stop pregnant dairy cows, acting as endocrine-active substances in promoting acne [47, 48].

A recent study involving 60 post-pubertal Chinese women with acne and/or hirsutism, a heterozygotic CYP21A2 mutation was reported in 8.3 % of the patients, significantly higher relative to the general population, but without significant differences in hormonal levels between the heterozygous carriers and normal subjects [49].

The ratio of the length of the index to ring fingers (2D:4D) is usually smaller in males compared with females. This has been suggested to reflect prenatal androgen exposure and sensitivity and shows an inverse correlation with prenatal testosterone levels. This ratio was found to be significantly lower in female acne patients aged between 16 and 36 years compared with healthy controls, and associated with upregulated cutaneous AR [50]. This should be investigated in children with neonatal or infantile acne.

Anti-androgen treatment in acne

A systematic review rated the use of spironolactone at doses ≤ 100 mg/day as low or very low for all outcomes based on the quality of evidence [51]. In addition to classical oral anti-androgens, topical anti-androgens have been investigated for acne treatment. Recent studies reported that topical clascoterone (cortexolone 17α propionate), a novel AR antagonist, has high AR binding affinity, inhibits AR-regulated transcription in a reporter cell line *in vitro*, and antagonizes androgen-regulated lipid and inflammatory cytokine production in a dose-dependent manner in primary human sebocytes [52]. Phase 3 randomized clinical trials demonstrated the effect and safety of clascoterone in acne treatment [53, 54]. However, the low success rate at twelve weeks (18.4/20.3 % with clascoterone vs. 9.0/6.5 % with the vehicle) raises concerns for its efficacy in clinical use. Increasing research effort has been made on nanotechnology to improve the topical drug delivery approaches of available anti-androgens, including spironolactone, flutamide, and cyproterone acetate, to enhance sustained delivery and deposition into the pilosebaceous units [55–57].

A crossover randomized controlled six-arm trial involving 200 women with PCOS indicated that COCs containing progestins with low androgenic or antiandrogenic activities (such as desogestrel, cyproterone acetate, and drospirenone) have similar effects on HA, whereas those with levonorgestrel are less effective on androgenic profiles and more harmful on the lipid status [58]. The therapeutic effects of COCs containing drospirenone or chlormadinone acetate are similar with regards to the clinical, metabolic, and hormonal improvement in PCOS [59].

There are four second-generation anti-androgens, namely abiraterone acetate, enzalutamide, apalutamide, and darolutamide, currently approved by the Food and Drug Administration in the US for treating metastatic castration-resistant prostate cancer [60]. Abiraterone is a derivative of steroidal progesterone that inhibits 17 α-hydroxylase/C17,20-lyase (CYP17A1); therefore, it represses DHT biosynthesis [61]. Its use in the treatment of HA associated with CAH is under clinical trial (ClinicalTrials.gov Identifier: NCT02574910). The selective androgen receptor modulators (SARMs) are small molecule drugs that serve as either AR agonists or antagonists. Like androgens, the SARMs enter the cytoplasm and bind to the AR, forming a SARM-AR complex that is translocated to the nucleus to act as a transcriptional regulator, and recruit cofactors and coregulatory proteins, modulating the transcriptional response to binding of the AR complex. The tissue-specific regulation, fewer side effects, and excellent oral and transdermal bioavailability make SARMs feasible options for current androgen-directed therapies [62].

Oral isotretinoin, although not antiandrogenic in pharmacology, has been documented to reduce the serum levels of testosterone, DHT, and prolactin in women with acne, as well as the free testosterone in women with PCOS and nodulocystic acne [63, 64]. In addition to its effect on reduction of the IGF-1 levels and AR expression [62], recent translational medicine studies indicate that isotretinoin upregulates the expression of p53, an inhibitor of AR expression, and interacts with c-myc-AR-p53 as a crucial regulatory network in sebaceous gland differentiation [65, 66]. Further studies should be conducted to provide more direct evidence in support of these observations.

Conclusions

Future studies on the pathogenetic roles of sex hormones in acne should investigate the intracrine activation of androgen precursors, including backdoor DHT biosynthesis, the 11-ketotestosterone, and its 5α-reduced metabolite in the activation of AR. At the molecular level, the non-genomic pathway and post-transcriptional regulation of gene expression, including the tissue-specific function and the cell type enriched miRNAs, are the novel checkpoints. Polycystic ovary syndrome and NCCAH remain as critical models for studying the effects of aberrant hormone levels on acne. The relevance of the findings involving normoandrogenemic patients of different ages has not been validated. A better understanding of comedogenesis regarding de-differentiation of sebocytes to keratinocytes will stimulate development of new treatment approaches. The interaction between hormone metabolism and skin microbiota is an interesting topic. Elucidation of the hormonal effect on inflammation may explain why elevated acne severity is predominantly seen in men. Breakthroughs in the development of topical antiandrogens and SARMs will provide new treatment options for acne.

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